

$$(T3) \quad f \sim p(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \theta > 0$$

$$n=3 \\ \tilde{\theta}_1 = \bar{X} = \frac{1}{3} \sum_{i=1}^3 X_i, \quad \tilde{\theta}_2 = X_{(2)}$$

$$M\xi = \int_0^{+\infty} x \frac{1}{\theta} e^{-x/\theta} dx$$

$$M\xi = - \int_0^{+\infty} x d(e^{-x/\theta}) = -x e^{-x/\theta} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x/\theta} dx = \theta$$

$$M\xi^2 = \int_0^{+\infty} x^2 \frac{1}{\theta} e^{-x/\theta} dx = 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

$$1) \tilde{\theta}_1 = \bar{X}$$

а) Несмещенность

$$M\tilde{\theta}_1 = M\left(\frac{1}{3} \sum_{i=1}^3 X_i\right) = \frac{1}{3} \sum_{i=1}^3 M X_i = \frac{1}{3} \cdot 3\theta = \theta$$

$\Rightarrow \tilde{\theta}_1 = \bar{X}$ — несмещ. оценка параметра θ .

б) Дисперсия

$$D\tilde{\theta}_1 = D\left(\frac{1}{3} \sum_{i=1}^3 X_i\right) = \frac{1}{9} \sum_{i=1}^3 D X_i = \frac{1}{9} \cdot 3\theta^2 = \frac{\theta^2}{3}$$

$$M\tilde{\theta}_1 = \theta \quad D\tilde{\theta}_1 = \frac{\theta^2}{3}$$

$$2) \tilde{\theta}_2 = X_{(2)}$$

Для $k \in \overline{n}$ плотность выхв.: $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F^{k-1}(x)(1-F(x))^{n-k} f(x)$

$$n=3, k=2$$

$$f_{X_{(2)}}(x) = 6 F(x)(1-F(x)) f(x), \quad x \geq 0$$

$$F(x) = 1 - e^{-x/\theta}, \quad f(x) = \frac{1}{\theta} e^{-x/\theta}$$

$$f_{X_{(2)}}(x) = 6(1 - e^{-x/\theta}) e^{-x/\theta} \frac{1}{\theta} e^{-x/\theta} = \frac{6}{\theta} (e^{-2x/\theta} - e^{-3x/\theta}), \quad x \geq 0$$

а) Несмещенность

$$M X_{(2)} = \int_0^{+\infty} x f_{X_{(2)}}(x) dx = \frac{6}{\theta} \int_0^{+\infty} x (e^{-2x/\theta} - e^{-3x/\theta}) dx$$

м.к. $\int_0^{+\infty} x e^{-ax} dx = \frac{1}{a^2}, a > 0$

$$M X_{(2)} = \frac{6}{\theta} \left(\frac{\theta^2}{4} - \frac{\theta^2}{9} \right) = 6\theta \left(\frac{1}{4} - \frac{1}{9} \right) = 6\theta \cdot \frac{5}{36} = \frac{5}{6}\theta$$

$\Rightarrow M \tilde{\theta}_2 = \frac{5}{6}\theta \neq \theta$, т.е. $\tilde{\theta}_2 = X_{(2)}$ - смещенная

$\tilde{\theta}_2^* = c X_{(2)}$ - исправл. оценка. Требуем несмещ.: $M \tilde{\theta}_2^* = c M X_{(2)} = c \cdot \frac{5}{6}\theta = \theta \Rightarrow c = \frac{6}{5} \Rightarrow \tilde{\theta}_2^* = \frac{6}{5} X_{(2)}$

б) Дисперсия исправл. оценки

~~на~~ $D X_{(2)}$.

$$X_{(2)} = X_{(1)} + (X_{(2)} - X_{(1)})$$

$X_{(1)} \sim \exp(\frac{3}{\theta})$, $X_{(2)} - X_{(1)} \sim \exp(\frac{2}{\theta})$ - и они независ.

Тогда

$$D X_{(2)} = D X_{(1)} + D(X_{(2)} - X_{(1)}) = \frac{\theta^2}{9} + \frac{\theta^2}{4} = \frac{13}{36}\theta^2$$

$$\Rightarrow D \tilde{\theta}_2^* = D \left(\frac{6}{5} X_{(2)} \right) = \frac{36}{25} D X_{(2)} = \frac{36}{25} \cdot \frac{13}{36} \theta^2 = \frac{13}{25} \theta^2$$

т.о. $M \tilde{\theta}_2^* = \frac{6}{5}\theta$ $D \tilde{\theta}_2^* = \frac{13}{25}\theta^2$

Среди несмещ. оценок наиболее экстр. та, у кот-ой дисперсия меньше

$$D \tilde{\theta}_1 = \frac{\theta^2}{3} < D \tilde{\theta}_2^* = \frac{13}{25}\theta^2$$

$\Rightarrow \tilde{\theta}_1 = \bar{X}$ более эффективна, чем $\tilde{\theta}_2^* = \frac{6}{5} X_{(2)}$

Человек с тачкой пер-ва Крамера-Рао

а) Проверка регулярности модели

Модель $p(x; \theta) = \frac{1}{\theta} e^{-x/\theta} I\{x \geq 0\}$, $\theta > 0$

1) $p(x; \theta)$ дифференцируема по θ ($\theta > 0$)

2) Носитель испр. не завис. от θ : $x \geq 0$

3) $\int_0^{+\infty} p(x; \theta) dx = 1$ -?

$$\frac{\partial}{\partial \theta} p(x; \theta) = \frac{\partial}{\partial \theta} \left(\frac{1}{\theta} e^{-x/\theta} \right) = e^{-x/\theta} \left(-\frac{1}{\theta^2} \right)$$

$$\text{Тогда } \int_{-\infty}^{+\infty} \frac{\partial}{\partial \theta} p(x; \theta) dx = \int_0^{+\infty} e^{-x/\theta} \left(-\frac{1}{\theta^2} \right) dx = -\frac{1}{\theta^2} \int_0^{+\infty} x e^{-x/\theta} dx -$$

$$\text{т.к. } \int_0^{+\infty} x e^{-x/\theta} dx = \theta^2, \int_0^{+\infty} e^{-x/\theta} dx = \theta, \text{ то } \int_{-\infty}^{+\infty} \frac{\partial}{\partial \theta} p(x; \theta) dx = -\frac{\theta^2}{\theta^2} - \frac{\theta}{\theta^2} = -\frac{1}{\theta} - \frac{1}{\theta} = -\frac{2}{\theta} \Rightarrow \text{модель регулярная}$$

5) Информационный пример

$$\ln p(x; \theta) = -\ln \theta - \frac{x}{\theta}, \text{ тогда } \frac{\partial}{\partial \theta} \ln p(x; \theta) = -\frac{1}{\theta} + \frac{x}{\theta^2}$$

$$\Rightarrow I(\theta) = M \left[\left(\frac{\partial}{\partial \theta} \ln p(x; \theta) \right)^2 \right] = M \left(-\frac{1}{\theta} + \frac{x}{\theta^2} \right)^2$$

$$I(\theta) = \frac{1}{\theta^4} M \xi^2 - \frac{2}{\theta^3} M \xi + \frac{1}{\theta^2}$$

$$(M \xi = \theta, M \xi^2 = 2\theta^2) \Rightarrow I(\theta) = \frac{2\theta^2}{\theta^4} - \frac{2\theta}{\theta^3} + \frac{1}{\theta^2} =$$

$$= \frac{2}{\theta^2} - \frac{2}{\theta^2} + \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

$$\text{Тогда } n=3: D\tilde{\theta} \geq \frac{1}{nI(\theta)} = \frac{1}{3 \cdot \frac{1}{\theta^2}} = \frac{\theta^2}{3}$$

Для $\tilde{\theta}_2 = \bar{X}$

$$D\tilde{\theta}_2 = \frac{\theta^2}{3} = \frac{1}{nI(\theta)} \Rightarrow \tilde{\theta}_2 = \bar{X} \text{ явл. эфф. оценкой}$$

Для $\tilde{\theta}_2^* = \frac{1}{3} X_{(2)}$ минимума дисперсии не достигается, ничего сказать не можем.

НО т.к. мы нашли эффективную оценку и эфф. оценка !, то $\tilde{\theta}_2^*$ не явл. эфф.

$$(T4) \quad p(x; \theta) = \begin{cases} \theta, & x \in (-1, 1), x \neq 0 \\ \frac{1}{2} - \theta, & x = 0 \\ \frac{1}{2} - \theta, & x = 2 \end{cases} \quad 0 < \theta < \frac{1}{2}$$

(из уст-а нормировки)

$$M\xi = \int_{-1}^1 x \theta dx + 0(\frac{1}{2} - \theta) + 2(\frac{1}{2} - \theta), \quad \int_{-\infty}^{+\infty} p(x; \theta) dx = \int_{-1}^1 \theta dx + 2(\frac{1}{2} - \theta) = 2\theta + 1 - 2\theta = 1$$

т.к. $\int_{-1}^1 x dx = 0$, то $M\xi = 1 - 2\theta$

по мом случайн. теор. и выбор 1^{st} м-мт.

$$\bar{X} = 1 - 2\theta \Rightarrow \hat{\theta}_2 = \frac{1 - \bar{X}}{2} \text{ - оценка ОММ}$$

Пусть m раз мы в $(-1, 1)$ и 0 . n -м раз мы в 0 или 2

$$L(\theta) = \theta^m (\frac{1}{2} - \theta)^{n-m}$$

$$\ln L(\theta) = m \ln \theta + (n-m) \ln(\frac{1}{2} - \theta)$$

$$\frac{d}{d\theta} \ln L(\theta) = \frac{m}{\theta} - \frac{n-m}{\frac{1}{2} - \theta}$$

$$\frac{m}{\theta} = \frac{n-m}{\frac{1}{2} - \theta} \Rightarrow m(\frac{1}{2} - \theta) = \theta(n-m)$$

$$\frac{m}{2} - m\theta = n\theta - m\theta \Rightarrow \frac{m}{2} = n\theta \Rightarrow \hat{\theta}_2 = \frac{m}{2n} \text{ - оценка ОММ}$$

$$\left(\frac{d^2}{d\theta^2} \ln L(\theta) \right) = -\frac{m}{\theta^2} - \frac{n-m}{(\frac{1}{2} - \theta)^2} < 0 \Rightarrow \text{эта м. гл.}$$

с) Оценка ОММ

$$\tilde{\theta}_2 = \frac{1 - \bar{X}}{2}$$

$$M\tilde{\theta}_2 = M\left(\frac{1 - \bar{X}}{2}\right) = \frac{1}{2} - \frac{1}{2} M\bar{X} = \frac{1}{2} - \frac{1}{2} M\xi$$

$$M\xi = 1 - 2\theta \Rightarrow M\tilde{\theta}_2 = \frac{1}{2} - \frac{1}{2}(1 - 2\theta) = \theta \Rightarrow \tilde{\theta}_2 \text{ несмещ.}$$

$$M\xi^2 = \int_{-1}^1 x^2 \theta dx + 0^2(\frac{1}{2} - \theta) + 2^2(\frac{1}{2} - \theta)$$

т.к. $\int_{-1}^1 x^2 dx = \frac{2}{3}$, то

$$M\xi^2 = \theta \cdot \frac{2}{3} + 4(\frac{1}{2} - \theta) = \frac{2}{3}\theta + 2 - 4\theta = 2 - \frac{10}{3}\theta$$

$$\Rightarrow D\xi = M\xi^2 - (M\xi)^2 = (2 - \frac{10}{3}\theta) - (1 - 2\theta)^2$$

$$D\xi = 2 - \frac{10}{3}\theta - 1 + 4\theta - 4\theta^2 = 1 + \frac{2}{3}\theta - 4\theta^2$$

$$\text{можн. } D\tilde{\theta}_1 = D\left(\frac{1}{2}\bar{X}\right) = \frac{1}{4} D\bar{X} = \frac{1}{4} \frac{D\xi}{n} = \frac{1 + \frac{1}{2} \cdot 0 + 4 \cdot 0^2}{4n}$$

$D\tilde{\theta}_1 \xrightarrow{n \rightarrow \infty} 0 \Rightarrow$ по гост. усл-ю $\tilde{\theta}_1$ - состоятельная

2) Оценка ДМТ

$$\tilde{\theta}_2 = \frac{m}{2n}$$

Для 1 накл. в-го поместим в центр 2-го:

$$P\{\xi \in (-1, 1), \xi \neq 0\} = \int_{-1}^1 \theta dx = 2\theta \Rightarrow m \sim B(n, 2\theta)$$

$$\mu m = 2n\theta, Dm = n \cdot 2\theta(1-2\theta)$$

$$\mu \tilde{\theta}_2 = \mu\left(\frac{m}{2n}\right) = \frac{1}{2n} \mu m = \frac{1}{2n} \cdot 2n\theta = \theta \Rightarrow$$

$\Rightarrow \tilde{\theta}_2$ несмещ.

$$D\tilde{\theta}_2 = D\left(\frac{m}{2n}\right) = \frac{1}{4n^2} Dm = \frac{1}{4n^2} \cdot n \cdot 2\theta(1-2\theta) = \frac{\theta(1-2\theta)}{2n}$$

$D\tilde{\theta}_2 \xrightarrow{n \rightarrow \infty} 0 \Rightarrow$ по гост. усл-ю $\tilde{\theta}_2$ - состоятельная

$$\text{Модель: } p(x, \theta) = \begin{cases} \theta, & x \in (-1, 1), x \neq 0 \\ \frac{1}{2} - \theta, & x = 0 \end{cases} \quad 0 < \theta < \frac{1}{2}$$

1) провер. по $\partial/\partial\theta > 0$
2) проверить норм. расп. на завис. от θ
 $\frac{\partial}{\partial\theta} \int p(x; \theta) dx = 0$ - ?

$$\frac{\partial}{\partial\theta} p(x; \theta) = \begin{cases} 1, & x \in (-1, 1), x \neq 0 \\ -1, & x = 0 \\ -1, & x = 2 \end{cases}$$

$$\text{Тогда } \int \frac{\partial}{\partial\theta} p(x; \theta) dx = \int_{-1}^1 1 dx - 1 - 1 = 2 - 1 - 1 = 0$$

$\Rightarrow$ модель регулярна

Упрощ. Рамера

Для 1 накл: если $x \in (-1, 1), x \neq 0$, то

$$\ln p(x; \theta) = \ln \theta, \quad \frac{\partial}{\partial\theta} \ln p(x; \theta) = \frac{1}{\theta}$$

если $x = 0$ или $x = 2$: $\ln p(x; \theta) = \ln\left(\frac{1}{2} - \theta\right)$,

$$\frac{\partial}{\partial\theta} \ln p(x; \theta) = \frac{1}{\frac{1}{2} - \theta}$$

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln p(x, \theta) \right)^2 \right], \text{ т.е. } I(\theta) = 2\theta \cdot \frac{1}{\theta^2} + (1-2\theta) \cdot$$

$$\frac{1}{(\frac{1}{2}-\theta)^2}, \text{ но } \frac{1}{2} - \theta = \frac{1-2\theta}{2}, \frac{1}{(\frac{1}{2}-\theta)^2} = \frac{4}{(1-2\theta)^2}$$

$$\Rightarrow I(\theta) = \frac{2}{\theta} + \frac{4}{1-2\theta} = \frac{2(1-2\theta) + 4\theta}{\theta(1-2\theta)} = \frac{2}{\theta(1-2\theta)}$$

$$\Rightarrow \text{если } V^* \text{ выборки } n, \text{ то } D\tilde{\theta} \geq \frac{1}{nI(\theta)} = \frac{\theta(1-2\theta)}{2n}$$

$$\text{Для } \tilde{\theta}_2 = \frac{m}{2n}$$

$$D\tilde{\theta}_2 = \frac{\theta(1-2\theta)}{2n}, \text{ т.е. } D\tilde{\theta}_2 = \frac{1}{nI(\theta)} \Rightarrow$$

$$\Rightarrow \tilde{\theta}_2 = \frac{m}{2n} - \text{эффективная оценка}$$

$$\text{Для } \tilde{\theta}_1 = \frac{1-\bar{X}}{2}$$

$$D\tilde{\theta}_1 = \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n}$$

$$\frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n} - \frac{\theta(1-2\theta)}{2n} = \frac{1 + \frac{2}{3}\theta - 4\theta^2 - 2\theta + 4\theta^2}{4n} =$$

$$= \frac{1 - \frac{4}{3}\theta}{4n}, \text{ т.к. } 0 < \theta < \frac{1}{2}, \text{ то } 1 - \frac{4}{3}\theta > 0 \Rightarrow$$

минимум не достигается, ~~и~~ и, следовательно, НД

т.к. эти оценки не совпадают, а эффективная оценка!, то $\tilde{\theta}_1$ не явл. эффективной